

# Jongha Kim

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<https://jonghakim35.github.io/>

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|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|
| EDUCATION          | <b>Korea University</b><br>M.S. & Ph.D. Integrated Student in Computer Science and Engineering<br>Advised by Professor Hyunwoo J. Kim at MLV Lab, KAIST<br>GPA : 4.15 / 4.5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Seoul, Republic of Korea<br>Sep 2022 - Current      |
|                    | <b>Korea University</b><br>B.S. in Computer Science and Engineering<br>B.S. in Statistics (Double Major)<br>GPA : 4.22 / 4.5 (Major : 4.25 / 4.5)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Seoul, Republic of Korea<br>Mar 2018 - Aug 2022     |
| RESEARCH INTERESTS | My goal is to develop personalized and reliable multimodal AI agents. My research focuses on personalizing Multimodal Large Language Models, exploring post-training methods ( <i>e.g.</i> , DPO), systems ( <i>e.g.</i> , RAG), and ways to leverage structured representations to complement these models to achieve the goal.                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                     |
| WORK EXPERIENCES   | <b>Adobe Research</b><br>• Research Scientist Intern<br>• Mentor: Dr. <a href="#">Joon-Young Lee</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | San Jose, CA, USA<br>May 2026 - Sep 2026 (Expected) |
|                    | <b>MLV Lab</b><br>• M.S. & Ph.D. Integrated Student at MLV Lab<br>• Advisor: Prof. <a href="#">Hyunwoo J. Kim</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Seoul, Republic of Korea<br>Sep 2022 - Current      |
|                    | <b>MLAI Lab</b><br>• Undergraduate Research Intern at MLAI Lab<br>• Advisor: Prof. <a href="#">Sung Ju Hwang</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Seoul, Republic of Korea<br>Jan 2021 - Jul 2022     |
|                    | <b>Upstage AI</b><br>• Teaching Assistant & Mentor of BoostCamp AI Tech 1-3rd course<br>• Collaborated with Prof. <a href="#">Tae Hyun Oh</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Seongnam, Republic of Korea<br>Dec 2020 - Jun 2022  |
|                    | <b>VoyagerX</b><br>• Developer Intern                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Seoul, Republic of Korea<br>Jul 2020 - Jan 2021     |
|                    | <b>SW Maestro</b><br>• 10'th trainee at SW Maestro Program                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Seoul, Republic of Korea<br>May 2019 - Nov 2019     |
|                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                     |
| PUBLICATIONS       | <p>[1] <b>DocPrune: Efficient Document Question Answering via Background, Question, and Comprehension-aware Token Pruning</b><br/>Joonmyung Choi, Sanghyeok Lee, <a href="#">Jongha Kim</a>, Sehyung Kim, Dohwan Ko, Jihyung Kil, Hyunwoo J. Kim<br/>In collaboration with <a href="#">Adobe</a> Research<br/>IEEE/CVF Conference on Computer Vision and Pattern Recognition (<b>CVPR</b>), 2026.</p> <p>[2] <b>Relevance-aware Multi-context Contrastive Decoding for Retrieval-augmented Visual Question Answering</b><br/><a href="#">Jongha Kim</a>, Byungoh Ko, Jeehye Na, Jinsung Yoon, Hyunwoo J. Kim<br/>In collaboration with <a href="#">Google</a> Cloud AI<br/>IEEE/CVF Conference on Winter Conference on Applications of Computer Vision (<b>WACV</b>), 2026 [paper]</p> |                                                     |

- [3] **TabFlash: Efficient Table Understanding with Progressive Question Conditioning and Token Focusing**  
Jongha Kim, Minseong Bae, Sanghyeok Lee, Jinsung Yoon, Hyunwoo J. Kim  
 In collaboration with [Google](#) Cloud AI  
 AAAI Conference on Artificial Intelligence (AAAI), 2026 [[paper](#)]
  - [4] **Improved Query Specialization for Transformer-based Visual Relationship Detection**  
Jongha Kim, Jihwan Park, Jinyoung Park, Jinyoung Kim, Sehyung Kim, Hyunwoo J. Kim  
 Information Sciences, 2026 (IF = 6.8) [[paper](#)]
  - [5] **VidChain: Chain-of-Tasks with Metric-based Direct Preference Optimization for Dense Video Captioning**  
 Ji Soo Lee\*, Jongha Kim\*, Jeehye Na, Jinyoung Park, Hyunwoo J. Kim  
 AAAI Conference on Artificial Intelligence (AAAI), 2025 [[paper](#)]
  - [6] **Groupwise Query Specialization and Quality-Aware Multi-Assignment for Transformer-based Visual Relationship Detection**  
Jongha Kim\*, Jihwan Park\*, Jinyoung Park\*, Jinyoung Kim, Sehyung Kim, Hyunwoo J. Kim  
 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2024 [[paper](#)]
  - [7] **Concept Bottleneck with Visual Concept Filtering for Explainable Medical Image Classification**  
 Injae Kim\*, Jongha Kim\*, Joonmyung Choi, Hyunwoo J. Kim  
 MedAGI Workshop at International Conference on Medical Image Computing and Computer-Assisted Intervention (MedAGI@MICCAI), 2023 (Oral Presentation) [[paper](#)]
  - [8] **Object Detection in Aerial Images with Uncertainty-Aware Graph Network**  
Jongha Kim, Jinheon Baek, Sung Ju Hwang  
 VOLI Workshop at European Conference on Computer Vision (VOLI@ECCV), 2022 [[paper](#)]
  - [9] **SuperClip Pyramid for Video Temporal Grounding**  
 Sanghyeok Lee, Juyeon Ko, Joonmyung Choi, Jongha Kim, Dohwan Ko, Ji Soo Lee, Hyunwoo J. Kim  
 In collaboration with [KT](#) Gen AI Lab  
 Under Review
  - [10] **Context-Calibrated Preference Optimization for Mitigating Object Hallucination in Multimodal Large Language Models**  
 Byungoh Ko, Jinyoung Park, Jongha Kim, Jeehye Na, Jaewon Cho, Hyunwoo J. Kim  
 Under Review
- (\* denotes equal contribution)

## PATENTS

### **Method and apparatus for dense video captioning based on chained task learning and metric loss for fine-grained understanding of video language model**

Hyunwoo Kim, Ji Soo Lee, Jongha Kim, Jeehye Na, Jinyoung Park  
 Korea Patent No.10-2025-0164550.

### **Information providing method and system for sharing fruit information including sugar content information measured through image vision processing**

Sanghoon Lee, Hyemin Song, Jongha Kim, Hyun Kim  
 Korea Patent No.10-2020-0153010.

### **Method for measuring sugar content of apple using image.**

Sanghoon Lee, Hyemin Song, Jongha Kim  
 Korea Patent No.10-2242155, registered on Apr 14, 2021

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|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|
| HONORS &<br>AWARDS   | <b>Travel Grant for CVPR 2024</b><br>CVPR 2024                                                                                                                                                        | Jun 2024                                   |
|                      | <b>Graduate School Outstanding Freshman Scholarship</b><br>Korea University                                                                                                                           | Sep 2022                                   |
|                      | <b>Dean's List</b><br>Korea University                                                                                                                                                                | Spring 2019                                |
|                      | <b>Semester High Honors</b><br>Korea University                                                                                                                                                       | Fall 2018/2019/2021, Spring 2019/2020/2021 |
|                      | <b>2020 Agrifood Public Big Data Startup Competition</b><br>Rural Development Administration                                                                                                          | Aug 2020                                   |
|                      | <b>National Scholarship for Science and Engineering</b><br>Korea Student Aid Foundation                                                                                                               | Spring 2018 - Spring 2022                  |
| ACADEMIC<br>SERVICES | Reviewer of <b>NeurIPS</b> (Conference on Neural Information Processing System)                                                                                                                       | 2024-2025                                  |
|                      | Reviewer of <b>CVPR</b> (IEEE/CVF Conference on Computer Vision and Pattern Recognition)                                                                                                              | 2024-2026                                  |
|                      | Reviewer of <b>ICLR</b> (International Conference on Learning Representations)                                                                                                                        | 2025-2026                                  |
|                      | Reviewer of <b>AAAI</b> (AAAI Conference on Artificial Intelligence)                                                                                                                                  | 2025-2026                                  |
|                      | Reviewer of <b>AISTATS</b> (International Conference on Artificial Intelligence and Statistics)                                                                                                       | 2025-2026                                  |
|                      | Reviewer of <b>ICML</b> (International Conference on Machine Learning)                                                                                                                                | 2025                                       |
|                      | Reviewer of <b>WACV</b> (IEEE/CVF Winter Conference on Applications of Computer Vision)                                                                                                               | 2026                                       |
|                      | Program Committee of <b>MedAGI@MICCAI</b> (MedAGI Workshop at International Conference on Medical Image Computing and Computer-Assisted Intervention)                                                 | 2024, 2025                                 |
| TALKS                | <b>Teaching Assistant</b><br>Special Topics in Artificial Intelligence (AAA740), Korea University                                                                                                     | Seoul, Republic of Korea<br>Fall 2024      |
|                      | <b>Korea University &amp; LG AI Workshop</b><br>Korea University                                                                                                                                      | Seoul, Republic of Korea<br>Feb 2023       |
|                      | <b>Defining and solving problems in deep learning projects</b><br>BoostCamp AI Tech 2nd course, Upstage AI                                                                                            | Virtual<br>Nov 2021                        |
|                      |                                                                                                                                                                                                       |                                            |
| SKILLS               | <b>Language</b>                                                                                                                                                                                       |                                            |
|                      | • Korean ( <i>native</i> )                                                                                                                                                                            |                                            |
|                      | • English ( <i>fluent</i> ), TEPS : 544/600 ( <i>officially described as "Native Level of English Proficiency"</i> )                                                                                  |                                            |
|                      | <b>Programming Skills</b>                                                                                                                                                                             |                                            |
|                      | • Python, PyTorch, Git, Bash (Most proficient)<br>• TensorFlow, SQL, Node.js, React.js, Vue.js (Experienced)<br>• Website built: <a href="#">MedAGI Workshop 2024</a> , <a href="#">Personal Blog</a> |                                            |

*Last Updated: Feb 21, 2026*